

Original Radiology Report

Findings 🧠:

- (1) Bilateral basal ganglia, corona radiata, and centrum semiovale show patchy T1/T2 hypointensity, FLAIR mixed signal, and DWI iso-signal.
- (2) Right basal ganglia shows long T1 and slightly long T2 signal.
- (3) Bilateral frontal subcortical and periventricular white matter show abnormal signals.
- (4) Ventricular system is mildly enlarged; sulci are deepened.

Impression 📄:

- (1) Multiple old lacunar infarcts in bilateral basal ganglia, corona radiata, and centrum semiovale.
- (2) Right basal ganglia encephalomalacia, likely old hemorrhage.
- (3) Demyelinating white-matter lesions; brain atrophy.



Structured Reports for Foundation Pretraining 📄

Disease category	Location	Modalities
Old lacunar infarct	Bilateral basal ganglia; corona radiata; centrum semiovale	T1, T2, FLAIR, DWI
Encephalomalacia / old hemorrhage	Right basal ganglia	T1, T2
White-matter demyelination	Bilateral frontal subcortical and periventricular white matter	T1, T2, FLAIR, DWI
Brain atrophy	Diffuse / global	-

(A)

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Findings 🧠:

- (1) A patchy lesion is located in the posterior horn of the right lateral ventricle and adjacent periventricular white matter, showing long T1 and long T2 abnormal signal.
- (2) FLAIR shows heterogeneous high signal, and DWI shows iso-to-low signal.
- (3) The lesion extends along the posterior horn of the right lateral ventricle, with obvious heterogeneous enhancement after contrast.
- (4) Mild surrounding edema is observed in the right periventricular region.

Impression 📄:

- (1) Space-occupying lesion centered around the posterior horn of the right lateral ventricle.
- (2) Right ventricular posterior horn involvement is considered.
- (3) Glioma is more likely.



Labels for Grounded Task Fine-tuning 📄

Subtype	Coarse-grained Localization	Fine-grained Localization
Glioma	Right lateral ventricle	Right ventricular posterior horn

Fine-grained Localization Vocabulary Normalization:

"Posterior horn of the right lateral ventricle"

"Right lateral ventricular posterior horn"

"Posterior horn region of the right lateral ventricle"

"Right ventricular posterior horn region"

"Right posterior horn of lateral ventricle"

} Right ventricular posterior horn

(B)